

What Brings You Here Today?: Challenges for Clinicians Communicating Data to Patients

Sarah Dunn
School of Informatics
Edinburgh, United Kingdom
s0678276@ed.ac.uk

Abstract

The growing availability of personal medical data presents new challenges for healthcare professionals tasked with communicating it to patients. Clinicians must frequently explain numerical data to patients in order to optimize patient outcomes, practice shared decision making and facilitate informed consent. However the challenges experienced by clinicians when communicating numerical medical data are not currently defined. Through the analysis of semi-structured interviews with 19 healthcare professionals I identify 12 challenges in communicating numerical healthcare data to patients and organize these around four themes: Data Properties, Clinician Characteristics, Patient Characteristics and Environmental Factors. I then describe in detail how these relate to the communication of numerical medical data by clinicians from a Human Computer Interaction (HCI) perspective. Finally, motivated by these challenges, I provide 3 opportunities and associated research questions for those in the HCI community to consider when working to improve the communication of numerical patient data.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Medical Data, Patient Communication, Challenges

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1 Research Problem and Motivation

Effective communication between healthcare professionals and patients is integral to the practice of medicine. Recent advances in medical technology and the increased emphasis on patient-centered care and shared decision making [25] have made it necessary for clinicians to explain complex, numerical, personal healthcare data to patients in a straightforward way [5]. This is challenging. In an attempt to tackle this problem, a variety of techniques involving

technology have been used to aid patient understanding. These include digital tools such as interactive online patient-decision aids [6, 15], large language models (LLMs) [49, 53], electronic health records (EHRs) [51] and novel data visualization techniques [5, 32, 33].

Patients often struggle to correctly understand and interpret information, including numerical medical data, due to low levels of health numeracy or literacy, a heightened emotional state or poor communication with their healthcare providers [2, 20]. Recently, the application of big data to healthcare [4] and increasing accessibility of medical data to patients [23, 46] has amplified this problem by increasing the volume and complexity of medical data available to patients. Patients are now exposed to data from continuous self-monitoring devices [28], electronic health records [23] home testing kits [48] and hospital monitor displays.

Clinicians are therefore tasked with explaining not only basic numerical concepts such as averages, arithmetic, and percentages, but also higher-level concepts such as estimation, probability, variability, and error metrics to patients [2]. Further examples of this include: identifying trends and thresholds when self-monitoring blood glucose values in diseases such as diabetes [28]; understanding proportions in terms of relative and absolute risks when making decisions about treatment [38] and explaining the accuracy and sensitivity of screening tests such as Prostate Specific Antigen (PSA) [3].

An inability to understand and interpret medical data impacts patient satisfaction and is associated with poorer health and medical outcomes for reasons including reduced treatment compliance and an impaired ability to make medical decisions [29, 38]. Thus, there is a growing need to improve patients' understanding of their personal medical data, and technology or techniques from HCI could potentially aid this. However, in order to do so, an appreciation of the specific challenges and subsequent requirements is needed.

My work contributes to this by identifying the challenges clinicians face when explaining numerical data to patients. In particular I focus on data communicated during day-to-day interactions in primary and secondary care, such as feedback from daily ward rounds in a hospital, a visit to a specialist at a clinic or an appointment with a primary care doctor (GP). In particular, I study the problems from the clinician's perspective, as the target users for any communication aids. This has never been done before with respect to explaining personal numerical medical data in day-to-day interactions.

To gain insight into the range of challenges clinicians face when communicating numerical data to patients, I conducted semi-structured interviews with clinicians from a broad range of specialties, and other clinical staff involved in communicating numerical data with

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patients. From this I identify 12 challenges, grouped into 4 main themes, namely: Data Properties, Patient Characteristics, Clinician Characteristics, and Environmental Factors. I then discuss how the identification of these challenges creates opportunities and questions for HCI researchers.

2 Related Work

Communicating medical information to patients in an understandable way has always presented a challenge for clinicians. However, as the nature of the information changes so too do the specific difficulties. Clinicians are now increasingly tasked with explaining personal, numerical healthcare data to patients. This is partly due to the shift from medical paternalism [14] towards approaches in which the patient has more autonomy, including shared decision making. Shared decision making involves the clinician and patient working together to make medical decisions aligned with the patient's values and preferences [26], and therefore relies on a patient understanding their personal data and its implications.

Insight into some of the challenges patients face with understanding healthcare data can be gained by studying previous work from the HCI and medical communities. There currently exist various guidelines alluding to the challenges involved with communicating numerical health data in public health campaigns [52] or for patients involved in research [5]. However, their relevance to challenges encountered when explaining an individual's personal numerical data during a face-to-face consultation is limited. Other guidelines have focused on extremely specific medical situations, such as the International Patient Decision Aid Standards guidelines (IPDAS) [22], a set of criteria proposed by the medical community for judging patient decision aids that may include data visualization [44]. Likewise, whilst tools produced by the HCI community to aid the communication of numerical data in health have proved successful e.g. [10, 12, 19, 34], these are applicable only to extremely specific medical scenarios, and findings may not be applicable for day-to-day clinical interactions.

Recent work of the most relevance by Rajabiyazdi *et al.*, identifies the challenges with patient clinician communication during day-to-day clinical appointments from the perspectives of both the clinician and the patient [34–37]. However, this work focuses primarily on the challenges associated with patients reporting health data they obtained via self-monitoring to a clinician during the diagnostic phase of the appointment, i.e. data they are already familiar with. Therefore, whilst it is likely their conclusions will be relevant, my research differs in a key aspect: I wish to identify the challenges *clinicians* face when communicating unfamiliar numerical data to a patient in the clinic.

3 Methods

As clinicians primarily learn communication skills through experience in the clinical environment, not textbooks [17], I thought it vital to consult with them firsthand when identifying their real-world, everyday challenges. Previous work has generally failed to involve clinicians directly, a problem also recognized by Rajabiyazdi *et al* [37], likely due to factors such as difficulty in obtaining interview time with clinicians. Therefore I considered it essential to prioritize speaking to a wide range of healthcare professionals who

must frequently communicate numerical data to patients. Through the use of mailing lists, networking at events, professional and personal contacts, I recruited 19 interviewees.

Interviewees were doctors (P1-P17), a research nurse (P18) and a dietitian (P19), all of whom work in patient-facing roles. Doctors from a wide range of specialties, including acute care, surgery, secondary care, psychiatry and primary care were recruited in order to capture the general challenges of communicating numerical data to patients, rather than specific challenges particular to one scenario. They had a wide range of experience: newly trained (<5 years qualified, $n = 4$), moderately experienced (6–20 years qualified, $n = 11$) and those with extensive experience (>21 years qualified, $n = 4$). This was motivated by the recent shift from paternalistic practices, to patient-centered care and shared decision making [21] which may affect the challenges encountered. All the doctors were based in the UK, working in the NHS, as was the research nurse. The dietitian was based in Switzerland. This study was approved by the University of Edinburgh's ethics review board, and all participants provided written informed consent.

The core interview questions were the same for all interviewees, based on insights from the literature combined with details I perceived as being of most relevance to HCI researchers. They covered four main topics: 1) Common scenarios in which clinicians discuss numerical data with patients 2) The specific needs and objectives of clinicians when communicating this data 3) Problems clinicians face when presenting numerical information to patients 4) Tools, techniques and approaches clinicians utilize to simplify numerical information for patients.

Interviews were conducted in person ($n = 5$) or online or via Microsoft Teams or Zoom ($n = 14$) and ranged from 30 minutes to an hour in length. All interviews were fully recorded and transcribed. For analysis of the interviews, I used a Template Analysis style of thematic analysis, as described by King *et al* [9]. Unlike Braun and Clarke's method of reflexive thematic analysis [7], Template Analysis develops initial themes (the "template") earlier in the process, and has a focus on iteratively developing and "trying out" the template during coding [24]. It does not make as much of a distinction between "codes" and "themes" [8]. Some preliminary codes were identified 'a priori' based on my review of the literature, for example, I hypothesized that 'Data complexity' would be one of the common problems; however I expected these to evolve during analysis. I aimed to identify the challenges associated with the holistic process of communicating numerical medical data, capturing aspects such as the type of data being communicated, the motivation when communicating data, the techniques used to make data relevant and interpretable to the patients, and the associated barriers and limitations.

The template was iteratively revised based on several discussions with my supervisory team. Similar codes were then grouped and organized into a hierarchical structure. The coding strategy was regularly discussed with my supervisory team and modified throughout. This approach allowed me to conduct the interviews in an efficient way to obtain the insights I sought, whilst affording the flexibility required to capture emerging themes in a largely unexplored field.

4 Results

This section discusses my final set of 12 challenges grouped into 4 challenge themes. I start with a detailed exploration of the challenges relating to **Data Properties**, as a key novelty of this research is its focus on the communication of personal numerical medical data, as opposed to medical information in general. I then briefly discuss the remaining themes in the order in which their implications inform each other: **Clinician Characteristics** and **Patient Characteristics** before finally addressing the **Environmental Factors** vital to consider when developing communication aids.

4.1 Challenge Theme: Data Properties (DP)

As this study is concerned with the communication of a patient's personal numerical medical data, I firstly outline the precise aspects of numerical medical data that create the greatest difficulties in patient understanding and recollection.

DP1: Data Interpretability. Numerical medical data can be hard to interpret due to its sheer **volume (DP1.1)**: “The information sheets, more simple ones might be 10 pages [...] commercial ones can be 35 pages long” [P18]. Medical conditions are often **multifactorial (DP1.2)**, demanding the assimilation of many independent data dimensions when making decisions: “...a lot of our statistics are written for the presumed healthy patient, but actually you might have a woman in front of you who's got a body mass index of 50 and COPD [chronic obstructive pulmonary disease]” [P14] [42]. This becomes even more complex when the different factors have **interdependencies (DP1.3)**, whereby changing one dynamically affects the others: “...complex clinical scenarios like heart failure when you're treating with diuretics to get rid of extra water, but diuretics impact kidneys and then normally in kidney damage we give more fluid, but you can't do that because you're trying to take the fluid away. And it's these kind of contradictions and things that can be more difficult to explain...” [P8].

However, difficulties interpreting numerical data also concern the **abstract concepts (DP1.4)** it portrays. This could be intangible notions like risk that patients find hard to relate to themselves “I find it [risk] abstract to interpret for myself, I suppose. These numbers don't have meaning quite a lot of the time...” [P9]. **Latent data (DP1.5)** can also pose challenges, for example conditions where despite medical data revealing problems, there are no immediate, discernible symptoms for the patient “...hypertension is another chronic condition where if you don't know what your blood pressure is doing, it doesn't have a symptomatic effect, but it has long term, very poor health outcomes and can cause multiple problems...” [P17]. Finally, whilst not a property of the data itself, much numerical data is communicated in the context of confusing **medical jargon (DP1.6)** “I often see people get too jargony and the patient being lost” [P17].

DP2: Data Uncertainty. Uncertainty is pervasive in medicine [21], and this is also true for medical data. This can relate to uncertainty over **future predictions (DP2.1)**, or prognoses, of patients: “...they asked, but it was virtually impossible to give them the risks and the odds. Head injuries vary so much...” [P3]. In some specialties, data is **missing or unknown (DP2.2)** sometimes due to administration issues, or lack of relevant research in that patients subgroup:

“...where data is missing, you know we can give kind our best guess about what we've seen previously but...” [P4]. **Ambiguous (DP2.3)** data and associated guidelines leads to patients expecting answers that clinicians can't give: “...a lot of people want to know if they're going through the menopause by the FSH blood test. And it's not as easy as doing a blood test and you get yes or no, you're in the menopause...” [P1]. Similarly, medical guidelines are constantly being updated based on new data, and **changing official medical advice (DP2.4)** has proved problematic “...everybody back in the 80s was on HRT. Then they found that actually there's an increase of breast cancer with HRT. So everybody was taken off it...” [P1]. Finally, **unexpected data (DP2.5)**, for example surprising test results, is a cause of patient confusion often requiring additional explanation “so that's one data point [where] people get a bit confused because they're like, oh, but I've already done that national screening, it was negative. And so you have to then explain, but actually that was only this cut off...” [P6].

DP3: Data Veracity and Value. Unsurprisingly, the majority of interviewees reflect on the growing problem of medical misinformation, in the form of **erroneous data (DP3.1)** caused by procedural or administrative errors. In particular the concepts of data **specificity and sensitivity (DP3.2)**: “...the perception of tests is also the fact that what the test says is 100 percent. And people don't realize you can get false negatives, or false positives and the harm that those can cause...” [P1]. As the volume of medical data increases, there is a potential for **data overload (DP3.3)** [27], so clinicians often don't pass on results they deem unimportant to patients “...you've got so many results coming through, if something comes back as normal, we just file it...” [P6].

DP4: Data Fallacies. Misinterpretation of medical data can be based on the patient's own **incorrect conclusions (DP4.1)**. Alternatively **misinformation from outside sources (DP4.2)** such as friends and relatives, the media or increasingly popular self-diagnostic tests often mean clinicians must first identify and understand the reasons for any incorrect thinking and correct it: “...you need to find out what it is that they've been using [as a source] and then try and find out about that. Of course, you only have ten minutes... you don't have very long to be doing a review of the literature that they've come in with...” [P10].

4.2 Challenge Theme: Clinician Characteristics (CC)

Clinicians are encouraged to reflect on their practice, and this is apparent from the interviews, with many giving candid insights into the challenges they experience themselves and observe in other medical professionals communicating numerical data to patients.

CC1: Clinicians' Abilities. Clinicians may struggle with an **inability to explain clinical data clearly (CC1.1)**, “And I've been on [ward] rounds where I [had] to go back and do another round and explain what the consultant said...” [P8]. Others reported **low data literacy (CC1.2)** for themselves or colleagues “...I don't have a particularly academic background. I have to remind myself of what statistics tell me and [...] how you then convey that to your patients in an understandable format...” [P10]. This may seem surprising, but it echoes findings in the literature [47].

CC2: Clinicians' Attitudes. Clinicians sometimes make **misjudgments (CC2.1)** as to how to pitch the data they share with patients leading to awkward situations: “...so then I had to kind of go back over what I'd said, but in much simpler terms...” [P4]. As corroborated by the literature [16], interviewees allude to **disagreements on data disclosure (CC2.2)** within or between treatment teams, due to differing beliefs about what a patient should know: “...I remember as a junior being told off by one of my consultants because I'd frightened one of her patients by talking about the risks...” [P14]. Some clinicians express a reluctance to use existing tools due to the **belief aids inhibit communication (CC3.3)** by detracting from the consultation: “sometimes any paper or anything external can actually get in the way of you talking to them [...] it almost becomes another barrier.” [P9].

4.3 Challenge Theme: Patient Characteristics (PC)

Patients are the recipients of the information in the exchange and so it is important to understand how an individual patients' characteristics influence the clinicians' decisions of how and when to communicate medical data.

PC1: Socioeconomic Factors. A patient's ability to consolidate and interpret the data may be limited long-term by **disabilities (PC1.1)**, e.g. learning disabilities, dementia: “... with poor memory they aren't able to retain information long enough to evaluate it...” [P13]. Patients with comprehensive medical histories, or prior knowledge from their occupation or education (e.g. former doctors) often possess **high health and data literacy (PC1.2)**. This presents a challenge to clinicians, as they don't wish to insult them “...there are also the patients who do understand very well and you don't want to patronize them...” [P17]. Many clinicians emphasized the difficulties related to **low health and data literacy (PC1.3)** within the general population c.f. [40, 45], for example when instructing patients who need to self monitor: “...you have to work out an average and a lot of my patients don't know how to work out the mean...” [P1]. **Language and cultural barriers (PC1.4)** were problematic, particularly in ethnically diverse communities, “...sometimes you don't have a sense that the translator is particularly data literate and so you know, you've no idea what they're saying...” [P14]. Finally, patients from **disadvantaged populations (PC1.5)** are less able to access or use their data, a finding mirrored in the literature [5]: “...people who really need healthcare are going to be deprived of it by affluent, articulate people who monopolise the resources [a diabetes management app] that are made available in every system...” [P10].

PC2: Willingness to Understand. Patients present to clinicians with a range of motivations and intentions. It is especially challenging if a patient arrives **pre-decided (PC2.1)**, and unwilling to act upon advice based on the data “...there are patients who no matter how many pictures [of the data] I showed them, will just never agree to take a statin...” [P1]. **Low patient interest (PC2.2)**, ranging from simple apathy to complete denial is difficult from a legal and ethical standpoint due to the need for informed consent [30]: “In many ways, the people who don't want to know or who are agnostic are a more challenging communication...” [P5]. Conversely, some patients become **fixated on the numbers (PC2.3)**, which can be

harmful: “you'll have these super anxious people, who will mull over every single bit of data...” [P1]. Lastly, patients may have **unrealistic expectations** and demand treatment not indicated by the data: “just because someone wants some procedure done doesn't necessarily mean that that's appropriate...” [P16].

PC3: Emotional Factors. Clinicians reflect that the **emotional state (PC3.1)** of a patient greatly impacts how they choose to communicate data: “The emotional state [...] makes a huge difference as to how much information somebody can retain and therefore how much is worth discussing with them” [P16], c.f. [13, 41]. The **personal impact of the data (PC3.2)** on the patient, be it delivering a terminal diagnosis or delaying their discharge, influences how the data is framed “...you don't want to pin hope on somebody like, this [test result] could get you home earlier for example...” [P2]. Finally, **Patient trust (PC3.3)** is paramount to communication c.f. [31], and communicating numerical data can either help or hinder this relationship: “...there'll be people who are still very upset... but when you give them space to talk about it and answer any questions, I think sometimes there is comfort in knowing this wasn't me [when communicating risks]...” [P14].

4.4 Challenge Theme: Environmental Factors (EF)

The environment in which clinicians share data with patients dictates the options available to them when communicating data, and the lack of basic resources is surprising.

EF1: Physical Environment. It may be chaotic and crowded and there is often a **lack of privacy (EF1.1)**, making communication of sensitive data problematic: “...space and privacy can be difficult...” [P8]. Some acute care clinicians have **no basic stationary (EF1.2)**, such as a pen and paper, which prevents them communicating data visually for example via sketches c.f. [13, 17]: “Can I get a bit of paper? Do I have a surface to write on?” [P17].

EF2: Technological Environment. In hospitals and patients' homes there can be **slow or unavailable internet access (EF2.1)**: “...people try and show things on patient websites and things don't load and they're a bit clunky...” [P9]. There may be **no access to hardware (EF2.2)** such as printers or screens to display information: “You don't usually have colour printers.” [P8]. In addition, multiple clinicians faced **difficulty using technology (EF2.3)** due to the different systems they had to negotiate: “you've got different [...] software wherever you go...” [P17]. This led to situations in which they were unaware of useful features for explaining numerical data, such as visualization tools: “...a lot of my colleagues don't use it, but it is there, maybe they don't know that it's there...” [P15, when referring to the use of Vision [41] an interface to display graphs of patients' test results]. This is one of several issues also highlighted in a recent report from the British Medical Association (BMA) regarding necessary improvements to the IT systems used in the NHS [18].

EF3: Time. As one clinician stated “Time's a big [challenge]”; **Short appointment times (EF3.1)** lead to situations where: “...we have 10 minutes to talk about some quite complicated information” [P2]. This can cause a reluctance to use communication aids that take up patient contact time for some clinicians, whilst others

shared an enthusiasm for anything that can “*communicate complex data more quickly*” [P15]. Unsurprisingly, **Medical emergencies (EF3.2)** allow the patient no time to process information: “*...from time of decision to delivery it has to be less than 30 minutes. So I’m rapidly barking that same form at these poor patients...*” [P14].

5 Opportunities and a Call to Action

This comprehensive set of challenges relating to the communication of data to patients by clinicians gives insight into current practice and open issues. My four themes, **Data Properties**, **Clinician Characteristics**, **Patient Characteristics** and **Environmental Factors** offer a lens for researchers to consider important factors when designing, implementing and evaluating communication tools. Although some challenges may appear quite general (e.g. **emotional state (PC3.1)**), I carefully outline their relevance to the HCI community via their impact on the specific problem of communicating numerical medical data. To conclude, I will use these challenges to discuss potential opportunities for HCI to aid clinicians in explaining numerical medical data to their patients.

5.1 Opportunity 1: Incorporate Flexibility (OP1)

The importance of tailoring communication to the situation, environment and needs of the clinician and patient is emphasized throughout the interviews. It is therefore vital to have flexibility built into communication aids in order for clinicians to use them successfully. For example, data visualizations could help reduce the cognitive load of assimilating the data and accommodate patients with **differing data literacy levels (PC1.2 and PC 1.3)**, **language and cultural barriers (PC1.4)** or those in **highly emotional states (PC3.1)** [24]. Communication tools must also adapt to the environment they are to be used in, with contingencies for any technological and practical constraints, e.g. **slow or unavailable internet access (EF2.1)** or **no access to hardware (EF2.2)**. Personalization of technology used by clinicians to communicate numerical data is currently an exciting and unexplored area. It offers huge potential, evidenced by recent work from the HCI community highlighting the benefits of tailoring data visualizations for patients with chronic conditions who track their medical data [34, 35], and interest in the personalization of healthcare information using automated comics [50]. However this barely scratches the surface, and poses the question of how a communication aids can be designed to accommodate all these different needs?

5.2 Opportunity 2: Design Responsibly (OP2)

“First do no harm” [43] is a mantra for clinicians pertinent to those working in patient data communication. The sharing of medical data has profound ethical and legal considerations. **Misinformation from outside sources (DP4.2)** negatively impacts patients’ receptiveness to the correct advice, and poor communication aids can lead to patients drawing **incorrect conclusions (DP4.1)** from the data. Exposing patients to unnecessary data can lead to **data overload (DP3.3)** and patients becoming overwhelmed. Before sharing data, it is important to consider the **personal impact of the data (PC3.2)**. In particular, **disadvantaged patient populations (PC1.5)** may be adversely affected, as has been seen with the introduction of tools such as My Diabetes My Way [11] and Data

Visualization resources [39] which may not be inclusive for individuals with low data literacy or from less-affluent demographic groups. HCI researchers should therefore apply guidance available from the medical community e.g. [22] on communicating to those with low health literacy. **Patient trust (PC3.3)** in clinicians is paramount. Thus any data communicated must be verified from reputable sources and frequently updated to reflect the **changing official medical advice (DP2.4)**. Finally, it is vital to rigorously evaluate the impacts of communication aids once they are deployed and devise strategies to mitigate these. However, how to do so remains an open question, also posed by the medical community [1].

5.3 Opportunity 3: Consider the Needs of the Communicator (OP3)

Clinicians may be skeptical about utilizing communication aids, due to a belief that **aids inhibit communication (CC3.3)** or a **difficulty using technology (EF2.3)** and should hence be involved at early stages in the design process. Any tools must be easy to use on account of challenges such as **short appointment times (EF3.1)**. There may be **disagreements on data disclosure (CC2.2)** and clinicians may make **initial misjudgments (CC2.1)** about how to pitch the data. Communication aids could alleviate these problems by adapting to allow clinicians to tailor and revise what data to share with patients and how to frame it, based on their current practices and perception of the patient. However this invites the question as to how we can involve clinicians, who may not have time for participatory design or focus groups in developing these aids?

6 Conclusion

Through my interviews I gain a first-hand account of the careful considerations clinicians make when communicating numerical data to patients. It is apparent that this requires an incredibly nuanced technique, perfected over years of experience and reflection, which cannot be learned from textbooks alone. Clinicians must achieve more than just clearly explaining the often confusing, complex and ambiguous numbers: They must then contextualize the potentially profound implications of these numbers for that individual in an empathetic, sensitive way, taking into account the patient’s history, preferences and current emotional state. Clinicians acknowledged that they are humans too, with their own limitations and biases, and this could impact how they approached communicating data. Finally, it is apparent that communication often takes place in far from ideal circumstances and environments, with limited time and resources.

Based on my findings, I present opportunities for HCI researchers to explore when developing communication aids for clinicians, highlighting the importance of flexible, responsible design that meets the individual needs of both the clinician and the patient. As one interviewee [P14] reflected: “*...in an ideal world [...] we’d approach this holistically and tailor all of this to the individual, but it’s just not always feasible in the current system*”. However, despite the challenges of the current system, clinicians strive to do the best for their patients. With that in mind, I invite the HCI community

to investigate the opportunities identified, in an effort to improve the current system and move slightly closer to an “ideal world”.

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